

DESIGN AND IMPLEMENTATION OF A BUCKLE-ACTIVATED IOT SMART HELMET FOR AUTONOMOUS ACCIDENT DETECTION AND EMERGENCY ALERTING

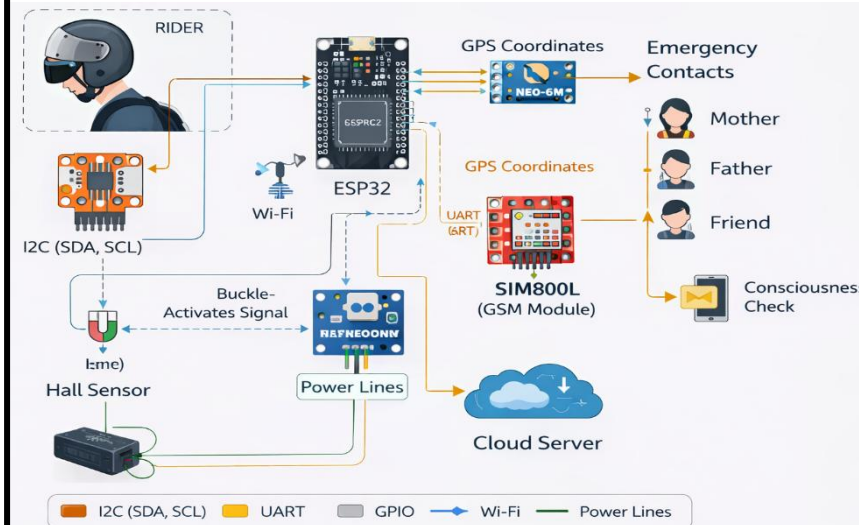


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OBJECTIVE

To design and develop a Smart Helmet System that enhances rider safety by integrating real-time monitoring and alert mechanisms. The system aims to detect accidents using motion sensors, track location via GPS, and automatically send emergency alerts through GSM communication. Additionally, it ensures rider compliance by verifying helmet usage and provides vibration-based warnings. The overall objective is to reduce fatalities and response time during road accidents using embedded systems and IoT-based technologies.

SYSTEM ARCHITECTURE



WORKING PRINCIPLE

The Smart Helmet operates using a combination of sensors and communication modules. The MPU6050 accelerometer and gyroscope detect sudden impacts or abnormal tilt, indicating a possible accident. The GPS module continuously retrieves location coordinates. When an accident is detected, the system triggers the GSM module to send an alert message with location details to be predefined contacts. A hall sensor ensures the helmet is worn, and a vibration sensor provides feedback alerts. The microcontroller processes all inputs and coordinates real-time responses efficiently.

EXPERIMENTAL SETUP

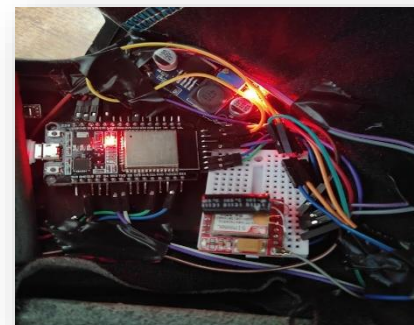
The system is built using a microcontroller (such as ESP32/Arduino), integrated with an MPU6050 sensor, GPS module, GSM module, hall sensor, and vibration motor. The components are interfaced through serial communication and GPIO pins. Power is supplied via a rechargeable battery. The helmet setup includes proper placement of sensors for accurate readings. Testing involves simulating accident conditions, verifying GPS accuracy, and ensuring timely SMS alerts. Serial monitoring is used for debugging and validating sensor outputs during real-time operation.



HALL SENSOR –
BUCKLE ACTIVATION

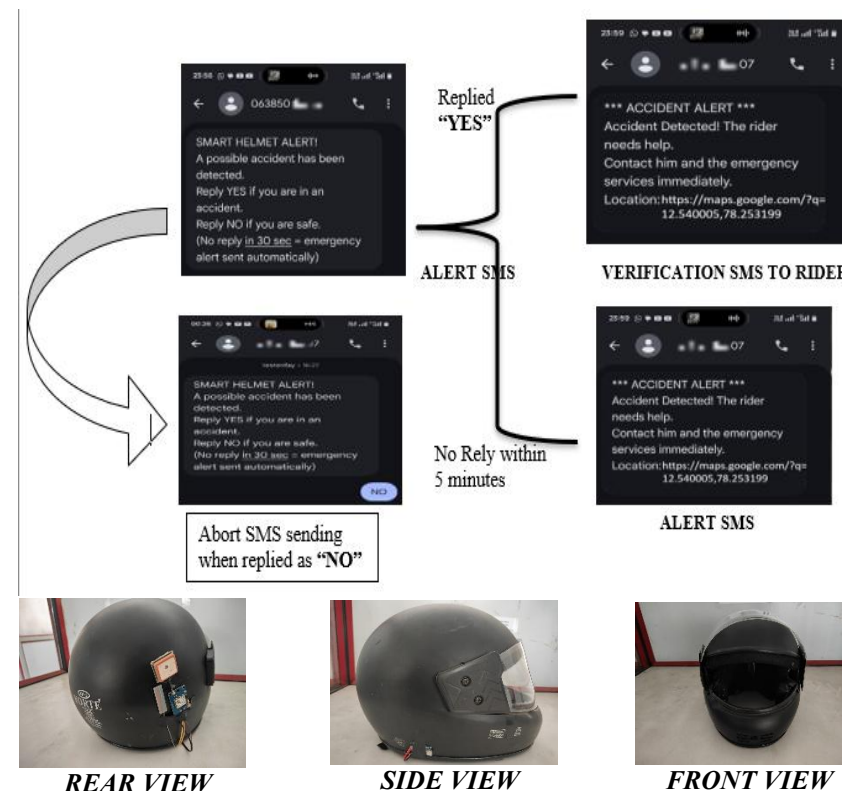


BATTERY MANAGEMENT SYSTEM
& POWER SUPPLY



MICROCONTROLLER & SENSORS
INSIDE HELMET VIEW

EXPERIMENTALS RESULTS



GUIDE: Dr. S. SUMATHI, Professor & HoD of ECE
BATCH No: 13

CONCLUSION

The Smart Helmet system successfully demonstrates an effective approach to improving road safety using embedded technology. It provides real-time accident detection, location tracking, and instant emergency communication. The system is cost-effective, scalable, and practical for real-world implementation. By reducing emergency response time and ensuring helmet compliance, it has the potential to significantly decrease accident-related fatalities. Future enhancements can include cloud integration, mobile applications, and advanced AI-based accident prediction for even greater reliability.